Leaper: next training runs

Eight full experiments + one short diagnostic | Planning document | 12 September 2026

Owner requirements

Keep the learning rate constant for the entire run. Required for ALL NINE experiments: **0.00015** from the first update through the last. Do not decrease it late in training, at checkpoints, or during an extension. Preserve this same rate on resume. Any future rate change requires an explicit new decision; no automatic decay schedule.

Keep all 34 observations unchanged. Leaper must learn body and leg clearance through experience. Do not add clearance sensors, change target-direction coordinates, or supply extra target information. Memory may process previous observations already available to the robot.

Keep static targets, randomized arenas and existing collision physics. Reverse is a separate action experiment. All new runs start from random weights; no pretrained walker is imported. This document does not launch training.

Why this batch

The completed LSTM reached 60.5% final arrival success on 200 development mazes, with a best checkpoint of 63.5%. Final failures were 68 stuck and 11 timeouts. Target memory persists after discovery, yet some recorded episodes moved without collisions and still failed to arrive.

The reward already favors successful arrival, but does not guarantee the policy learns the required actions. Weak time costs, distance feedback, future-reward weighting, recovery actions and memory handling are separate explanations to test. None is established as the sole cause.

Common experiment settings

Setting	Contract
Architecture	Existing additive stateful LSTM, except the explicit-window comparison.
Learning rate	Constant 0.00015 throughout, including later training and extensions.
Budget	Runs 1-8: 2,031,616 transitions each. Run 9: initial 106,496-transition diagnostic.
PPO settings	8 environments x 1,024 steps = 8,192 transitions per rollout; batch 256; 10 epochs. Gamma 0.995 except run 7. Other existing PPO settings remain fixed.
Evaluation	Approximately every 50,000 transitions, rounded to rollout boundaries; final evaluation included. Same 200 development mazes; one training seed per configuration.

The run list

ALL RUNS: CONSTANT LEARNING RATE 0.00015.
No decay, no late reduction, and no reduction on resume or extension. This applies to the control, every comparison and the short diagnostic.

Each row changes only the stated settings relative to run 1. Reward cost values below are positive magnitudes subtracted from reward. Time-cost changes apply only after target discovery; search rewards remain unchanged.

Run	Exact change	Question
1. Control	**LR: constant 0.00015.** Existing forward-only LSTM. Time cost 0.002; distance coefficient 0.2; gamma 0.995.	Reference under the constant learning rate.
2. Reverse	**LR: constant 0.00015.** Signed throttle: negative reverses, zero stops, positive moves forward.	Does backward movement improve recovery and arrival?
3. Moderate urgency	**LR: constant 0.00015.** Post-discovery time cost 0.01 per step.	Does stronger time pressure reduce moving timeouts?
4. Stronger urgency	**LR: constant 0.00015.** Post-discovery time cost 0.05 per step.	Does more pressure help, or increase collisions and early failures?
5. Distance feedback	**LR: constant 0.00015.** Post-discovery distance coefficient 0.25: reward approach and penalize retreat equally.	Does 25% stronger distance feedback improve pursuit?
6. Reverse + urgency	**LR: constant 0.00015.** Combine run 2 reverse with run 3 time cost of 0.01.	Do recovery freedom and urgency work better together?
7. Longer reward horizon	**LR: constant 0.00015.** Gamma 0.999 instead of 0.995.	Does stronger weighting of distant outcomes improve pursuit?
8. Explicit memory window	**LR: constant 0.00015.** 64 consecutive existing observations; reset recurrent state for each window. Match training and inference.	Does finite-window processing improve behavior compared with carried state?
9. Aggressive diagnostic	**LR: constant 0.00015.** Post-discovery time cost 0.25; initial 106,496-transition budget.	Does very strong time pressure improve behavior or encourage early failure?

Priority and concurrency

Highest priority: runs 1-5. All configurations are independent enough to run simultaneously, subject to a measured CPU/RAM throughput check. With two effective slots, queue pairs 1/2, 3/4, 5/6 and 7/8; schedule diagnostic 9 in the next available slot. Do not assume nine concurrent jobs are faster.

Run 9 is a behavior and stability screen. Extend it only if arrival and failure behavior justify a larger declared budget. A low early score alone does not establish its long-term potential. Preserve all completed and interrupted results.

What the reward changes mean

Distance coefficient: 0.2 to 0.25

Distance reward = coefficient x (distance before movement - distance after movement). This uses straight-line world distance after target discovery. It is not a flat penalty per backward step. Time, collision and arrival rewards are separate.

Movement result	Current: 0.2	Proposed: 0.25
1 unit closer	+0.20	+0.25
1 unit farther away	-0.20	-0.25
Same distance	0	0

Moving away and recovering the same distance cancels before time costs and discounting. Stronger shaping amplifies both approach and retreat feedback; it does not directly punish circling at constant distance. A maximum-speed ten-step retreat covers at most 3.75 units: current distance cost 0.75, proposed 0.9375, plus time costs.

Discount factor: 0.995 to 0.999

Value now = reward now + gamma x next reward + gamma squared x the following reward + ... Higher gamma gives both future rewards and future penalties greater weight. The actual arrival payout remains +25.

Arrival reward timing	Gamma 0.995	Gamma 0.999
Immediately	25.00	25.00
100 steps later	15.14	22.62
500 steps later	2.04	15.16

The table isolates the arrival reward; route costs also contribute. Higher gamma may help connect earlier detour actions with eventual success, but can make value learning harder. It also reduces the discount advantage of arriving sooner, so time costs remain relevant.

Time cost: urgency rather than clearance information

Post-discovery cost per step	Cost of 100 steps	Cost of 500 steps
0.002 (control)	0.2	1
0.01	1	5
0.05	5	25
0.25 (diagnostic)	25	125

These are undiscounted time costs only. The 0.25 diagnostic is 125 times the current rate and can change the task incentive substantially. Measure whether the robot starts ending episodes early rather than reaching the target.

Execution and decision rules

Before launching

1. Recover and verify the intended seeker code revision: the handoff documents were read from Git commit 7689aca because they were absent from the current checkout. Preserve existing workspace changes and archive source hashes for every run.

2. Validate constant optimizer learning rate at the first, middle and final updates and after checkpoint resume. Verify no schedule, callback or extension can silently lower it.

3. Validate reverse mapping, reward arithmetic and window boundaries. Window inputs must not cross workers or episodes; padding must not become fake experience. Confirm the original 34-feature observation contract remains unchanged.

4. Benchmark concurrent throughput, including evaluations. Keep each run in a separate artifact folder. Record model counts, seeds, packages, initial weights and empty optimizer state.

5. Add separate actor/critic and memory/direct-branch gradient diagnostics and actual weight-update measurements. Verify logging does not change optimizer updates.

Measure outcomes, not reward alone

Primary measure: unassisted target arrival. Also record hidden-only discovery; post-discovery arrival and timeouts; remaining target distance at timeout; stuck/frozen failures by discovery phase; collision rates; pursuit duration with failures included; reverse usage; and reward components separately before and after discovery.

Compare runs at equal training budgets. Report best and final checkpoints separately. A lower collision rate, a shorter failed episode or a higher shaped reward is not enough to claim improvement. The one-seed-per-configuration design screens hypotheses; it does not establish repeatability across training seeds.

Keep the 1,000 reserved confirmation mazes (120000-120999) untouched until candidate and control checkpoints are frozen. Development mazes remain 30000-30199. Do not replace the browser production model during this research batch.

Launch and progress reporting

Announce each launch with its hypothesis, architecture/parameter count, exact changes, random initialization, budget and constant learning rate. Provide five-minute updates during active training. No launch or monitoring restart has been performed for this PDF request.

Evidence and scope

Plan based on the owner decisions in this conversation; SEEKER_HANDOFF.md, SEEKER_SCRATCH_RUN_LOG_2026-09-09.md and SEEKER_FINDINGS_AND_NEXT_EXPERIMENTS_2026-09-10.md recovered from commit 7689aca; and saved campaign artifacts under rl_artifacts/seeker_scratch_lr_20260909/.

Key evidence: residual_lstm_extension_2m/dev_2031616_summary.json; dev_2007040_episodes.json; gradient_audit/audit.json. The current reward and action definitions were checked in rl_environment.py at the documented revision. New configurations above are prospective, not measured improvements.